Guided Notes: Random Forests and Variance Reduction — Complete

Completed responses are marked "Answer."

Time: about 45–50 minutes. **Before this lesson:** decision-tree splits, variance, covariance, and bootstrap sampling. We move from one tree to bagging and then to random forests. The derivation concerns **regression predictions at a fixed input** x; classification is discussed after the derivation.

Learning goals

1. Explain why averaging trees is attractive and why bootstrap samples alone may leave their predictions correlated.
2. Derive the variance of an average of correlated tree predictions and interpret its limiting value.
3. Explain how random feature selection can lower covariance, and why a smaller feature set is not automatically better.
4. Use out-of-bag (OOB) predictions to assess a forest without using its training predictions as a performance estimate.

1. Motivation: one tree, many possible fits

A deep decision tree can change substantially when the training observations change. A different sample can produce a different top split and a different prediction at the same x. This is **high prediction variance**. If several trees make errors in different directions, averaging their predictions can make the result more stable.

For tree $b = 1, \ldots, M$, write $T_b(x)$ for its prediction and

$$\bar{T}(x) = \frac{1}{M} \sum_{b=1}^{M} T_b(x).$$

1A. Four regression trees predict $8, 12, 10, 14$ at the same x. Compute their average. If all four instead predicted 11, would the average change? Which set offers more potential for cancellation of errors around 11?

$$\bar{T}(x) = \frac{8 + 12 + 10 + 14}{4} = 11.$$

Answer. Four predictions of 11 also average to 11. The first set has errors on both sides of 11, so averaging can cancel those differences. Four identical predictions offer no such cancellation. This does not tell us which set is closer to the unknown true response.

From averaging to bagging

We have one training set, not M independent training sets. **Bagging** creates M bootstrap samples: for each tree, draw n observations *with replacement* from the n training observations, fit a tree, then average its prediction with the others. The bootstrap samples overlap, so their fitted trees need not be independent. For classification, a common aggregation rule is majority vote.

1B. Why do different bootstrap samples help? Why might they still give similar trees if one predictor is especially strong?

Answer. Different bootstrap samples can change the fitted splits and predictions. But the samples overlap, and a very strong predictor can be chosen near the root of nearly every tree. Their predictions may therefore remain correlated.

2. How much does averaging reduce variance?

Consider repeated runs of the training procedure, including new training samples and its random choices. At a fixed x, treat each $T_b(x)$ as a random variable. To keep notation short, write T_b and $\bar{T}$. The true target function $f(x)$ is fixed here; it is not itself a random covariance term.

If the trees were independent and each had variance σ^2, then

$$\mathrm{Var}(\bar{T}) = \mathrm{Var}\left(\frac{1}{M}\sum_{b=1}^{M} T_b\right) = \frac{1}{M^2}\sum_{b=1}^{M}\mathrm{Var}(T_b).$$

2A. Complete the independent-tree result. What happens as M grows?

$$\mathrm{Var}(\bar{T}) = \frac{M\sigma^2}{M^2} = \frac{\sigma^2}{M}.$$

Answer. Under independence, variance approaches 0 as M increases. Bootstrap trees are generally not independent, so this is a benchmark, not the usual forest result.

Bootstrap trees share training observations and often choose similar splits. For correlated trees, the exact identity is

$$\mathrm{Var}(\bar{T}) = \frac{1}{M^2}\left[\sum_{b=1}^{M}\mathrm{Var}(T_b) + \sum_{b\neq c}\mathrm{Cov}(T_b, T_c)\right].$$

For a simple model, assume equal tree variance $\mathrm{Var}(T_b) = \sigma^2$ and equal pairwise correlation $\mathrm{Corr}(T_b, T_c) = \rho$ for $b \neq c$. Thus $\mathrm{Cov}(T_b, T_c) = \rho\sigma^2$. This is a model for understanding averaging, not a promise that real forests have identical pairwise correlations.

2B. Count the variance terms and the **ordered** covariance terms in the identity. Substitute the counts and simplify, showing the intermediate line.

$$\begin{aligned}\mathrm{Var}(\bar{T}) &= \frac{1}{M^2}\left[M\sigma^2 + M(M-1)\rho\sigma^2\right] \\ &= \frac{\sigma^2}{M} + \frac{M-1}{M}\rho\sigma^2 \\ &= \rho\sigma^2 + \frac{1-\rho}{M}\sigma^2.\end{aligned}$$

Answer. There are M variance terms and $M(M-1)$ ordered covariance terms: each of M trees pairs with each of the other $M-1$ trees.

2C. Evaluate the last expression for $\rho = 0$, $\rho = 1$, and $M \to \infty$ (with σ^2 and ρ held fixed). What does the limit say about adding trees?

Case	$\mathrm{Var}(\bar{T})$	**Meaning**
$\rho = 0$	σ^2/M	Uncorrelated predictions get the full $1/M$ reduction.
$\rho = 1$	σ^2	Perfectly correlated, equal-variance predictions gain nothing from averaging.
$M \to \infty$	$\rho\sigma^2$	With fixed positive ρ, the shared covariance remains.

Answer. More trees shrink the $\sigma^2(1-\rho)/M$ part but cannot remove the $\rho\sigma^2$ part under this model. If $\rho = 0$, the limit is 0.

Variance-minimization lever. Write $c = \rho\sigma^2$ for the common pairwise covariance. Then

$$\mathrm{Var}(\bar{T}) = \frac{\sigma^2}{M} + \frac{M-1}{M}c.$$

For fixed $M > 1$ and fixed individual-tree variance σ^2, lowering c lowers ensemble variance. Adding trees shrinks the first term, while changing the split candidates can shrink the second. This is the mathematical motivation for random feature selection; its actual effect must be measured because changing m can also change σ^2 and bias.

2D. Suppose $\sigma^2 = 9$ and $\rho = 1/3$. Compute the variance for $M = 4$ and $M = 20$, and the limiting variance. How much does increasing M from 4 to 20 remove, and what remains?

$$\begin{aligned}
\mathrm{Var}(\bar{T}_4) &= 9\left(\frac{1}{3} + \frac{2/3}{4}\right) = 4.5,\\
\mathrm{Var}(\bar{T}_{20}) &= 9\left(\frac{1}{3} + \frac{2/3}{20}\right) = 3.3,\\
\lim_{M \to \infty} \mathrm{Var}(\bar{T}_M) &= 9\left(\frac{1}{3}\right) = 3.
\end{aligned}$$

Answer. Going from 4 to 20 trees removes $4.5 - 3.3 = 1.2$ variance units; 3.3 remains at 20 trees, and even infinitely many trees retain the floor of 3 in this simplified model.

3. Random features attack the shared part

Bagging changes which observations each tree sees. A **random forest** also changes which predictors a tree may use: at *each split*, it draws m of the p predictors as candidates and chooses the best split among those m. A fresh subset is drawn at the next split. Setting $m = p$ gives bagged trees under the same tree-growing rules.

When one feature dominates, bagged trees often use it near the root. Similar early splits can make their predictions move together. Excluding it from some candidate sets forces other split paths and can lower the covariance between tree predictions.

3A. Suppose $p = 10$, $m = 3$, and there is one dominant predictor. At a particular split, what is the probability it is in the candidate set? What is the probability it is excluded? Would reducing m make exclusion more or less likely?

$$P(\text{available}) = \frac{m}{p} = \frac{3}{10} = 0.30, \qquad P(\text{excluded}) = 1 - \frac{m}{p} = \frac{7}{10} = 0.70.$$

Answer. Reducing m makes exclusion more likely at a given split. This probability concerns one specified predictor and one random candidate draw; it does not say that the predictor appears in exactly 30% of all fitted trees.

3B. Follow the causal chain: smaller m → candidate sets → split choices → covariance → ensemble variance. Where could this strategy hurt prediction quality?

Answer. Smaller m more often removes the dominant predictor from the candidate set, causing some trees to take alternative split paths. Their predictions may become less correlated, lowering the shared covariance term. If useful features are excluded too often, individual trees may predict less accurately; their variance and the forest's bias may also change. The net effect must be checked empirically.

The **correlation floor** $\rho\sigma^2$ applies to the simplified equal-correlation model when these quantities remain fixed as M increases. It is the common pairwise covariance in that model. In real tuning, changing m can change **both** ρ and σ^2, as well as bias. Smaller m does not guarantee a lower floor or lower test error. The often-used $m \approx \sqrt{p}$ (especially for classification) is a starting heuristic, not a mathematical optimum.

3C. Two hypothetical settings have the same $M = 20$ and similar bias. Compute their ensemble variances. Which setting is preferable *on variance alone*?

Setting	σ^2	ρ	$\rho\sigma^2 + (1-\rho)\sigma^2/M$
A: more candidate features	9	0.40	$9(0.40 + 0.60/20) = 3.87$
B: fewer candidate features	12	0.15	$12(0.15 + 0.85/20) = 2.31$

Answer. B has lower ensemble variance even though its individual trees have larger variance. The decrease in correlation more than offsets that increase in this example. A and B were stipulated to have similar bias; without that stipulation, this table alone cannot rank test MSE.

For squared-error regression at a fixed x, expected test error also includes squared bias and irreducible noise:

$$\mathrm{MSE}(x) = \mathrm{Bias}(x)^2 + \mathrm{Var}(\bar{T}(x)) + \sigma_\varepsilon^2.$$

3D. Why can setting B have lower ensemble variance yet higher test MSE? Why is the variance formula above not literally a formula for majority-vote classification error?

Answer. B could have substantially larger squared bias if its trees frequently miss useful predictors, enough to outweigh its $3.87 - 2.31 = 1.56$ variance advantage. The displayed formula concerns an average of numeric regression predictions. Majority vote is a discrete rule, and its misclassification probability depends on the joint distribution of the votes and the true class; it is not given by this variance formula. Averaged class probabilities are numeric and can have a variance decomposition, but that variance alone does not equal classification error.

4. Evaluation: did the extra randomness help?

For any one tree, a training observation is **out of bag (OOB)** if it was never drawn into that tree's bootstrap sample. To predict observation i for OOB evaluation, aggregate only the trees for which i was OOB. Compare those predictions with the observed responses using an appropriate metric. Keep any final test set separate from this training and tuning process.

4A. With n draws from n training observations, what is the probability that one specified observation is OOB for one tree? Give the formula and its large-n approximation.

$$P(\text{OOB}) = \left(1 - \frac{1}{n}\right)^n \approx e^{-1} \approx 0.368.$$

Answer. One draw misses the specified observation with probability $1 - 1/n$; all n draws miss it with probability $(1 - 1/n)^n$. Thus about 36.8% of training observations are OOB for a given tree when n is large.

4B. You fit forests with $m = 3$ and $m = 10$. Their training errors are 0.01 and 0.00; their OOB errors are 0.18 and 0.24. Which m has better evidence of generalization, and why? What additional comparison and final check would you make?

Answer. $m = 3$ has the lower OOB error (0.18 versus 0.24), so it has better evidence of generalization on these data. Training errors are optimistic because they evaluate fitted trees on data used to grow them. Compare both forests with a simple baseline using the same metric, and evaluate the selected setting once on an untouched test set. Feature preparation must also avoid using information from the test set.

Exit check

E1. In one sentence, state why adding many bagged trees can stop giving much variance reduction.

Answer. If bagged trees make correlated errors, the covariance term remains as M grows, so adding trees alone eventually has little effect on variance.

E2. In one sentence, state what random feature selection changes and the tradeoff it introduces.

Answer. Random feature selection changes the split candidates at each node and can reduce tree covariance, but too small an m can weaken the trees or increase bias.

Sources for this lesson: `notes/RandomForestVariance.md`; STAT 486 *Random Forests* lecture, slides 2–10 (`github/slides/14-random-forest.pptx`); James, Witten, Hastie, Tibshirani, and Taylor, *An Introduction to Statistical Learning: With Applications in Python* (2023), §§8.2.1–8.2.2, pp. 343–347. The variance exercises and numerical examples are constructed for these notes.